Conceptual Model (Class Diagram)

Prepared per Lab 2 Manual, Section 3.2 (Object Model) — initial version, to be finalized in Lab 3

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1. Overview

The conceptual model below was derived from the Lab 1 Use Case Descriptions and Data Dictionary terms. Every Data Dictionary term became either an entity class or a documented simplification: Frame and Screen Capture were treated as transient in-memory data rather than persisted classes (per NFR-4, no raw video is retained), and UnimportantContentRegion / Scroll Trigger Zone were folded into attributes/behavior of ImportantContentRegion and AutoscrollController respectively rather than modeled as their own classes. Objects are grouped into the standard entity/boundary/control stereotypes (Bruegge & Dutoit) and laid out in three columns matching that grouping in the diagram below.

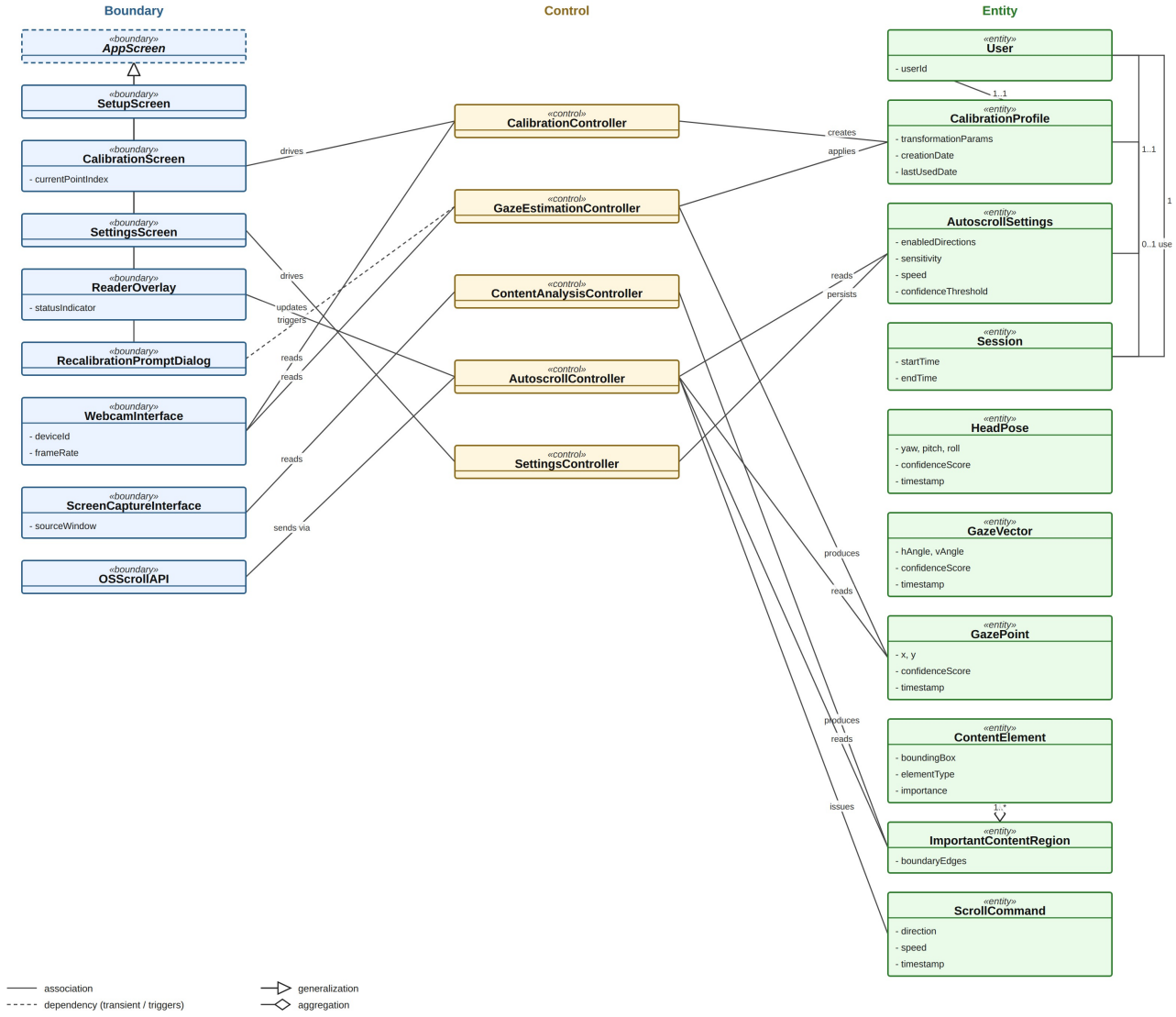


Figure 1. Doomscroll conceptual model: boundary, control, and entity classes with association, generalization, and aggregation relationships.

2. Key Boundary and Control Classes

The nine boundary classes correspond directly to the five Lab 1 UI mockups (SetupScreen, CalibrationScreen, SettingsScreen, ReaderOverlay, RecalibrationPromptDialog) plus the three external-system boundaries the app talks to (WebcamInterface, ScreenCaptureInterface, OSScrollAPI); AppScreen is an abstract generalization of the five screen/overlay/dialog classes, capturing their shared lifecycle. The five control classes each own one of the app's functional pipelines end-to-end — calibration, gaze estimation, content analysis, autoscroll decision-making, and settings persistence — and are the classes most likely to map onto the eventual controller/service layer of the implementation.

3. Class List

All 24 classes, grouped by stereotype, with a one-line statement of responsibility for each.

BOUNDARY CLASSES		
AppScreen (abstract)	*«boundary»*	Abstract base for all top-level application screens; defines shared show/hide/lifecycle behavior common to the five concrete screens below.
SetupScreen	*«boundary»*	First-run screen that requests camera permission and orients the user before calibration (Lab 1 mockup s1).
CalibrationScreen	*«boundary»*	Displays the sequence of on-screen calibration points and collects gaze samples at each (Lab 1 mockup s2).
SettingsScreen	*«boundary»*	Autoscroll settings panel: sensitivity, speed, per-direction enable/disable, confidence threshold (Lab 1 mockup s3).
ReaderOverlay	*«boundary»*	Always-on-top HUD shown during active reading; displays the autoscroll status indicator (Lab 1 mockup s4).
RecalibrationPromptDialog	*«boundary»*	Modal dialog offering to recalibrate when gaze-tracking confidence drops (Lab 1 mockup s5).
WebcamInterface	*«boundary»*	Boundary to the physical webcam device; exposes the continuous raw video frame stream.
ScreenCaptureInterface	*«boundary»*	Boundary to the OS screen-capture API; supplies raw on-screen content for classification.
OSScrollAPI	*«boundary»*	Boundary to OS-level scroll injection (PyObjC / pywin32) that issues scroll events to the focused window.

CONTROL CLASSES		
CalibrationController	*«control»*	Drives the calibration point sequence and builds a CalibrationProfile from the collected head-pose/gaze samples.
GazeEstimationController	*«control»*	Runs the continuous head-pose → gaze-point estimation pipeline and triggers the recalibration prompt on sustained low confidence.
ContentAnalysisController	*«control»*	Classifies captured screen content into important vs. unimportant regions.
AutoscrollController	*«control»*	Combines the current GazePoint and ImportantContentRegion(s) to decide when/how to scroll, and issues ScrollCommands accordingly.
SettingsController	*«control»*	Reads user input from the Settings screen and persists it as AutoscrollSettings.

ENTITY CLASSES

User	*«entity»*	The person using Doomscroll; associated with one CalibrationProfile, one AutoscrollSettings, and one or more Sessions.
CalibrationProfile	*«entity»*	Learned regression parameters mapping head-pose/gaze features to screen coordinates, plus creation/last-used dates.
AutoscrollSettings	*«entity»*	User-configurable sensitivity, speed, per-direction enable/disable flags, and confidence threshold.
Session	*«entity»*	One period of app usage (start/end time), relating a User's calibration and settings use over time.
HeadPose	*«entity»*	Estimated yaw/pitch/roll of the user's head at a point in time, with a confidence score.
GazeVector	*«entity»*	Estimated eye-in-head gaze direction at a point in time, with a confidence score.
GazePoint	*«entity»*	Estimated screen-space (x, y) gaze location, derived from HeadPose + GazeVector via the CalibrationProfile.
ContentElement	*«entity»*	One classified piece of on-screen content: bounding box, element type, and importance.
ImportantContentRegion	*«entity»*	An aggregated region of 1..* “important” ContentElements currently visible on screen.
ScrollCommand	*«entity»*	A direction/speed instruction issued to the OS scroll API.

Note: per the Lab 2 manual, this conceptual model is an initial version and is expected to be refined further in Lab 3 as the design is elaborated.